# When Do Kolmogorov–Arnold Networks Help in Wireless? A Matched-Capacity, Non-Saturated Benchmark

Phuc Hao Do

***Abstract*—Kolmogorov–Arnold networks (KANs) are increasingly proposed for wireless tasks with two claims: they match or beat multilayer perceptrons (MLPs) at fewer parameters, and their learned univariate edge functions make them interpretable. Rather than accept or dismiss these claims, we map *when* they hold, using a parameter-matched, multi-seed protocol on three non-saturated wireless regimes: power-amplifier behavioral modeling, OFDM channel estimation, and automatic modulation classification on the real RML2016.10a dataset. The honest picture is regime-dependent. KANs are genuinely competitive and sometimes better: a spline-KAN beats a matched MLP on modulation classification above about $2$ dB SNR and across the full SNR range, a Fourier-KAN with a richer basis beats the MLP on the regression task at matched capacity, and KANs win in the low-data regime. But the two most common KAN variants never dominate: a small, task-appropriate convolutional network beats every flat model on classification by a wide margin at a fraction of the parameters, at the default budget a plain MLP is the stronger flat model at low SNR, and a properly built classical model ties or beats the KANs on regression. An interpretability audit shows the KAN's edges are simple (a cubic fits them with median $R^2 = 0.998$), yet the symbolic extraction that would make them interpretable costs over a decibel, so the readable model is not the performant one. We conclude that the sweeping KAN "efficiency wins" reported for wireless track unmatched capacity and favorable benchmarks, give a regime map of where KANs actually help, and release a reusable protocol and harness.**

***Index Terms*—Kolmogorov–Arnold networks, wireless machine learning, modulation classification, digital predistortion, model interpretability, benchmarking.**

## I. Introduction

KOLMOGOROV–Arnold networks (KANs) [1] replace the fixed node nonlinearities of a multilayer perceptron (MLP) with *learnable* univariate functions on the edges, parameterized by splines [1] or truncated Fourier series [2]. Two properties are repeatedly advertised: a favorable accuracy-per-parameter trade-off against MLPs, and interpretability, because the learned one-dimensional edge functions can in principle be read off and replaced by closed-form symbols. Within roughly a year these claims have propagated into wireless communications: KANs have been proposed for modulation recognition, channel estimation, digital predistortion (DPD) [3], radio-frequency fingerprinting [4], and 6G signal intelligence [5].

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The wireless literature, however, has adopted KANs largely in a *showcase* mode: a KAN is shown to beat some baseline on some task, and the advantage is attributed to the architecture. Two methodological problems make such claims hard to trust, and both are known in the broader machine-learning community [6]. First, **capacity is rarely matched**: a small KAN is compared against a differently sized MLP, so "fewer parameters" can be an artifact of the comparison rather than a property of the model. Second, **benchmarks are often saturated**: when every model already exceeds, say, $99\%$ accuracy, differences are within noise and the comparison cannot distinguish architectures at all. A recent survey of KANs [7] and a fairness study on vision and tabular tasks [6] both caution that KAN advantages shrink or vanish once these controls are applied, but the question has not been asked cleanly *for wireless*, where the tasks, inductive biases, and baselines differ.

This paper asks the question directly and answers it with measurement. We do not propose a new architecture; we propose an honest evaluation and report what it shows, whichever way it falls. Concretely:

- We define a **parameter-matched, multi-seed benchmarking protocol** (Section III) in which KANs and baselines are sized to the same parameter budget, trained identically, and compared with bootstrap confidence intervals and paired differences. Baselines are chosen to be *strong and task-appropriate*, not strawmen.
- We evaluate three wireless regimes chosen to be **non-saturated**, so differences are resolvable: (i) power-amplifier behavioral modeling / DPD, a smooth low-dimensional regression that is the regime most favorable to KANs (Section IV); (ii) automatic modulation classification (AMC) on the real RML2016.10a dataset (Section V); and (iii) OFDM channel estimation, an estimation task with a strong LMMSE baseline (Section VI).
- We conduct an **interpretability audit** (Section VII) that separates two things a claim of interpretability must satisfy: are the learned edges simple, and is the simple form free? We find the edges are simple but the symbolic distillation that realizes the interpretability claim degrades accuracy.
- We run **ablations** (Section VIII) that stress-test the comparison in KAN's favor: parameter-budget, *symmetric* KAN-and-MLP hyperparameter, training-set-size, and SNR sweeps. These reveal honest crossovers, KANs win

in some regimes, and pin down where.
- We audit **computational cost** (Section IX): at matched parameters, KANs are 4 to 30× slower per inference than the MLP, so the "efficiency" that motivates them is largely illusory once compute, not just parameter count, is measured.
- We synthesize a **regime map** and practical guidance (Section X): KANs are competitive and win in specific regimes (high-SNR classification, low data, richer bases) but never dominate, are beaten by the right inductive bias, and carry a measurable interpretability cost. We release all code (Section XI).

Our aim is neither to promote nor to dismiss KANs, but to replace showcase claims with a regime map: to show precisely where the imported value propositions hold and where they do not, and to give the community a protocol that surfaces this before a claim is believed.

## II. Background and Related Work

### A. Kolmogorov–Arnold networks

KANs are motivated by the Kolmogorov–Arnold representation theorem [8], which states that any multivariate continuous function can be written as a finite composition of continuous univariate functions and addition. Where the classical universal-approximation results for MLPs [9], [10] place the nonlinearity at the nodes and learn linear edge weights, KANs [1], [11] invert this: the edges carry learnable univariate functions and the nodes merely sum. A KAN layer maps $\mathbf{x} \in \mathbb{R}^{d_{\text{in}}}$ to $\mathbf{y} \in \mathbb{R}^{d_{\text{out}}}$ through a learnable univariate function on each edge:

$$y_j = \sum_{i=1}^{d_{\text{in}}} \phi_{ji}(x_i) + b_j, \tag{1}$$

where each $\phi_{ji}$ is a spline [1] or a truncated Fourier series [2]. The spline variant we use follows the standard efficient formulation [2]: $\phi_{ji}(x) = w^b_{ji}\,\text{silu}(x) + \sum_c w^s_{jic} B_c(x)$, with $B_c$ the B-spline basis on a fixed grid. The Fourier variant uses $\phi_{ji}(x) = \sum_{k=1}^{K} \left[a_{jik}\cos(kx) + b_{jik}\sin(kx)\right]$. Both give a per-layer parameter count of the form $d_{\text{in}} d_{\text{out}}\, C$, where $C$ is the per-edge coefficient count; this is the quantity we hold fixed when we speak of a matched budget. A rapidly growing family of variants swaps the edge basis (wavelets in Wav-KAN [12], Gaussian radial bases in FastKAN [13], Chebyshev polynomials in Cheby-KAN [14]), and recent work also examines the hardware inference cost of these choices [15]; none changes the parameter accounting above, so our matched-budget comparison of the two most common instantiations, spline and Fourier, is representative. We use PyTorch implementations [16] following the efficient spline formulation [2].

### B. KAN claims and their audits

The original proposal [1] emphasizes accuracy-per-parameter on scientific regression and symbolic-regression interpretability. Subsequent scrutiny is mixed: [6] finds that under a fair comparison KANs beat MLPs mainly on symbolic formula tasks and often lose on vision and tabular data, and a broad survey [7] catalogues the fast-growing but largely showcase-style application literature. The interpretability claim in particular is usually asserted qualitatively (a plot of a few edges) rather than tested for fidelity. This matters because interpretability in machine learning is a substantive property, not a byproduct of a readable plot: the literature distinguishes genuinely interpretable models from post-hoc explanations of black boxes [17], [18], [19], and the question of whether a simpler surrogate preserves a model's behavior is exactly the subject of knowledge distillation and model-compression studies [20], [21]. Our fidelity audit (Section VII) applies that lens to KAN's symbolic-extraction claim.

### C. KANs in wireless

Deep learning is now pervasive in the wireless physical layer [22], [23], and KAN adoption there is recent and application-driven. For DPD, several KAN and symbolic-KAN models have been proposed to linearize power amplifiers with reduced complexity [3], [24], [25]; for radio-frequency fingerprinting, KAN-augmented residual networks report identification gains [4]; and signal-intelligence surveys for 6G list KANs among emerging tools alongside large language models [5], [26]. Across these works, the comparison is typically against an unmatched MLP or a single baseline, and the benchmark is not checked for saturation. The tasks themselves are well studied independently: AMC on the RML2016 datasets [27], [28], [29] is a standard classification benchmark, best attacked with convolutional inductive bias [30], whose low-SNR band remains far from saturation [31], [32]; PA behavioral modeling has a mature classical baseline in the memory polynomial and its generalizations [33], [34], [35]; and richer real-world benchmarks such as DeepSense 6G [36] confirm that many wireless tasks are far from solved. Interpretable-AI work for the physical layer [37] motivates the interpretability question but does not quantify KAN fidelity. Our contribution is to place KANs on these tasks under a matched-capacity, non-saturated, multi-seed protocol and to audit interpretability quantitatively.

## III. Benchmarking Protocol

The protocol (Fig. 1) exists to remove the two confounds identified in Section I. It has four rules.

*(1) Matched capacity.* For each task we fix a target parameter budget $P$ and size every model to it. For a two-layer network with input width $d$, hidden width $h$, and output width $c$, the parameter counts are

$$P_{\text{MLP}} = dh + h^2 + hc + (2h + c), \tag{2}$$

$$P^{\text{F}}_{\text{KAN}} = 2K\,(dh + hc) + (h + c), \tag{3}$$

$$P^{\text{S}}_{\text{KAN}} = (1{+}G{+}k)\,(dh + hc), \tag{4}$$

where $K$ is the Fourier order and $(G, k)$ the spline grid and degree. The trendy factor $2K$ or $(1{+}G{+}k)$ multiplies the same $dh{+}hc$ connectivity, so a KAN and an MLP reach the same $P$ only at different hidden widths. We fix the Fourier-KAN, compute its $P$ from (3), and solve (2) and (4) for the

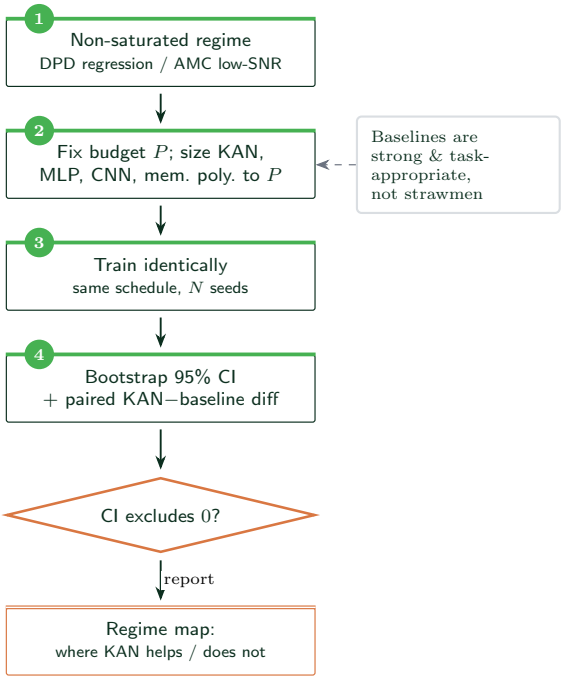



Fig. 1. The honest benchmarking protocol: fix a parameter budget, size every model to it, train identically over multiple seeds, and report bootstrap intervals and paired differences on a non-saturated regime. Baselines are strong and task-appropriate.

MLP and spline-KAN widths that land within a few percent of it. This makes "KAN uses fewer parameters" testable rather than assumed. For DPD all learned models sit at $P \approx 2330$ parameters; the residual mismatch is under $3\%$.

*(2) Strong, task-appropriate baselines.* A KAN must be compared against the baseline a practitioner would actually use. For the smooth regression regime that is a memory polynomial fit by regularized least squares [34], the classical DPD workhorse; we condition its basis so it is not a strawman. For the structured classification regime it is a small convolutional network that exploits the I/Q time structure. The matched MLP is the fairness control common to both.

*(3) Multiple seeds and interval estimates.* Each configuration is trained over several seeds (five for the regression regime, ten for classification); throughout, $\pm$ denotes one standard deviation across seeds, and bracketed intervals are bootstrap $95\%$ confidence intervals of the mean. For the head-to-head question we report the *paired* KAN-minus-baseline difference with its own bootstrap interval, and call a difference significant only when that interval excludes zero. No result rests on a single seed.

*(4) Non-saturated regimes.* We deliberately choose operating points where no model is near the ceiling, so a real difference can appear. This is the opposite of the common practice of reporting on a band where every model exceeds $99\%$.

Both KAN variants, the MLP, the CNN, and the memory-polynomial baseline are implemented in a single harness so that training schedule, optimizer, and data pipeline are identical across models; only the architecture changes. All neural models are trained with the Adam optimizer [38] under a cosine learning-rate schedule and identical epoch budgets, and the memory-polynomial baseline is fit by regularized least squares. Table I summarizes the models and how each is sized to the shared budget.

TABLE I
MODELS AND MATCHED-CAPACITY SIZING. EACH LEARNED MODEL IS SIZED TO THE SHARED PER-TASK BUDGET $P$; THE CNN IS INCLUDED AS THE STRONG STRUCTURAL BASELINE FOR AMC.

| Model | Edge / layer function | Sized by |
|---|---|---|
| KAN (spline) | B-spline on each edge | hidden width |
| KAN (Fourier) | truncated Fourier per edge | width, order $K$ |
| MLP | linear + ReLU nodes | hidden width |
| CNN | 1-D conv. over I/Q | fixed (small) |
| mem. poly. | $x\|x\|^{k-1}$ basis, LS fit | memory, order |

## IV. Regime 1: PA Behavioral Modeling (DPD)

### A. Task and data

Power-amplifier (PA) behavioral modeling is a smooth, low-dimensional nonlinear regression and is the regime KANs are *claimed* to handle best, since each edge can in principle capture a one-dimensional AM/AM-like curve. We drive a Wiener–Hammerstein PA with a pulse-shaped 16-QAM baseband signal at 6 dB input backoff. With input FIR $h_1$, output FIR $h_2$, and $u[n] = (h_1 * x)[n]$, the static Saleh nonlinearity [35] applies amplitude and phase distortion

$$A(r) = \frac{\alpha_a r}{1+\beta_a r^2}, \qquad \Phi(r) = \frac{\alpha_p r^2}{1+\beta_p r^2}, \tag{5}$$

so that the PA output is

$$y[n] = \Big(h_2 * v\Big)[n], \quad v[n] = A(|u[n]|)\, e^{\,j\left(\angle u[n] + \Phi(|u[n]|)\right)}. \tag{6}$$

Because (5)–(6) is not a memory polynomial, the classical memory-polynomial baseline gets no unfair "true model class" advantage; all learners approximate. The task is to predict the complex PA output $y[n]$ from a length-$M{=}5$ memory window of the complex input, and the metric throughout the paper is the normalized mean-squared error

$$\mathrm{NMSE} = 10\log_{10}\frac{\sum_n \|\hat{y}[n]-y[n]\|^2}{\sum_n \|y[n]\|^2}\ \text{[dB]}. \tag{7}$$

Models are trained over 5 seeds. We evaluate *behavioral modeling*, the forward-model accuracy that a predistorter is built on, rather than a full indirect-learning DPD loop; we therefore report NMSE and not ACPR or EVM, and read the results as a statement about learning a smooth PA nonlinearity, not about end-to-end linearization. Because the ground-truth PA is a Saleh model, its AM/AM curve is smooth by construction; a measured PA with strong memory or hysteresis could shift the boundaries, as we note in Section X.

### B. Results

Table II and Fig. 2 report the outcome. At a matched budget of 2330 parameters, the plain MLP achieves the best NMSE, $-16.27$ dB, beating the Fourier-KAN ($-15.5$ dB) and the

TABLE II
DPD / PA MODELING, MATCHED CAPACITY, 5 SEEDS. NMSE IN DB (LOWER IS BETTER).

| Model | Params | NMSE [dB] ↓ |
|---|---|---|
| MLP (matched) | 2354 | $\mathbf{-16.27 \pm 0.12}$ |
| GMP (classical) | – | $-16.20 \pm 0.05$ |
| KAN, Fourier | 2330 | $-15.50 \pm 0.06$ |
| KAN, spline (canon.) | 2280 | $-14.09 \pm 0.08$ |

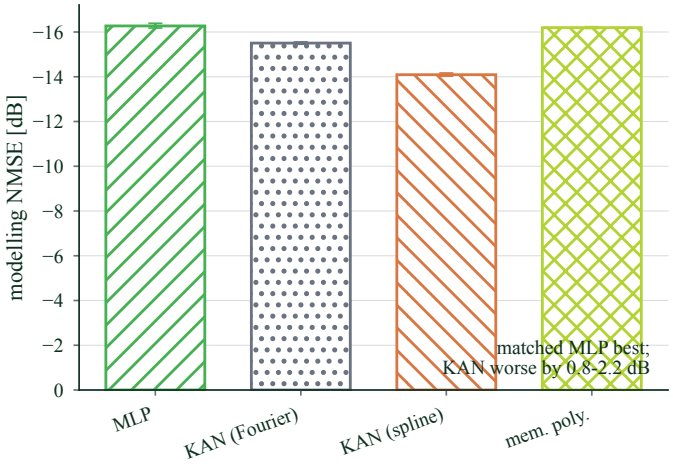



Fig. 2. DPD regime, matched capacity. A same-budget MLP achieves the best modeling NMSE; both KAN variants are worse (paired differences 0.8–2.2 dB, intervals exclude zero).

TABLE III
AMC ON RML2016.10A, LOW-SNR BAND $[-6, 6]$ DB, MATCHED FLAT-MODEL CAPACITY, 10 SEEDS. ACCURACY (HIGHER IS BETTER).

| Model | Params | Accuracy ↑ |
|---|---|---|
| CNN (task-appropriate) | 5291 | $\mathbf{0.658 \pm 0.005}$ |
| MLP (matched) | 68651 | $0.495 \pm 0.003$ |
| KAN, spline (canon.) | 69420 | $0.468 \pm 0.006$ |
| KAN, Fourier | 68395 | $0.349 \pm 0.007$ |

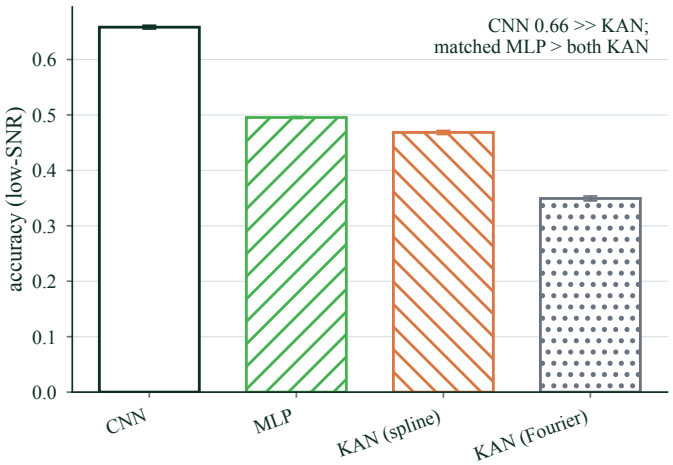



Fig. 3. AMC regime (real RML2016.10a, low-SNR). A small CNN dominates; among matched flat models the MLP beats both KANs. The correct inductive bias, not the edge parameterization, decides the task.

canonical spline-KAN ($-14.09$ dB). The paired differences are $+0.77$ dB (Fourier-KAN worse than MLP) and $+2.18$ dB (spline-KAN worse than MLP); both confidence intervals exclude zero, so the MLP advantage is significant. Crucially, the classical baseline is not a strawman: a properly built generalized memory polynomial (GMP) [34],

$$\hat{y}[n] = \sum_{k \text{ odd}} \sum_{m} \sum_{|m'-m| \le 2} c_{k,m,m'}\, x[n-m]\, \big|x[n-m']\big|^{k-1}, \tag{8}$$

with the lagging/leading cross-envelope terms $m' \neq m$ that a diagonal memory polynomial omits, reaches $-16.2$ dB, statistically tied with the best neural model and better than both KANs. So at the *default* matched configuration, even on the regime most favorable to KANs, the KANs beat neither a same-budget MLP nor the right classical method, and the widely used spline-KAN is the worst option of all. This ranking is not the whole story, however: Section VIII shows that a Fourier-KAN with a richer basis overtakes the MLP at matched capacity, and that KANs win when training data is scarce. The point of the default result is only that KAN superiority is not automatic; it must be earned and measured.

## V. REGIME 2: MODULATION CLASSIFICATION (RML2016.10A)

### A. Task and data

Automatic modulation classification is a high-dimensional, structured-input classification task. We use the real RML2016.10a dataset [28], which contains 128-sample complex I/Q frames over 11 modulations and a range of SNRs. We restrict to the *low-SNR band* from $-6$ to 6 dB, where accuracy is far from saturation and model differences are resolvable, and draw 40000 power-normalized frames. The flat models see the 256-dimensional flattened I/Q vector; the CNN sees the $2 \times 128$ frame. Models are trained over 10 seeds.

### B. Results

Table III and Fig. 3 report the outcome. The small convolutional baseline, with only 5291 parameters, reaches 0.658 accuracy and dominates every flat model. Among the flat models at a matched budget of about 68k parameters, the MLP (0.495) again beats both KANs; the spline-KAN (0.468) is close but below, and the Fourier-KAN (0.349) is markedly worse. The paired Fourier-KAN-minus-MLP difference is $-0.146$ (KAN worse), and the CNN leads the best KAN by $+0.309$; both intervals exclude zero. One lesson is unconditional and mirrors nothing in the regression regime: the right inductive bias, a convolution over the I/Q sequence, beats all flat models by a wide margin at a fraction of the parameters. The MLP-over-KAN ordering, however, is specific to this low-SNR band, and it inverts elsewhere: the SNR sweep in Section VIII shows the spline-KAN overtaking the MLP above about 2 dB and across the full SNR range, so our choice of the hardest band, made to avoid saturation, happens to be where the MLP looks best. This is exactly why an honest benchmark must resolve the operating point rather than report a single aggregate.

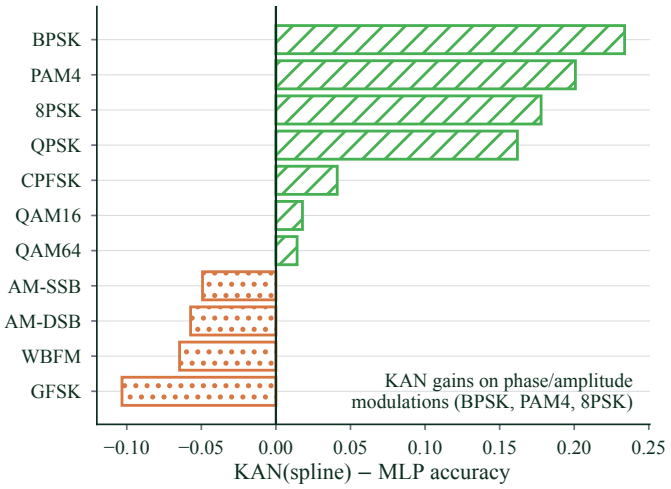



Fig. 4. Per-modulation accuracy difference, spline-KAN minus matched MLP, on the mid-band (0 to 12 dB) where the KAN leads overall. The KAN advantage concentrates on constellation-geometry modulations (BPSK, PAM4, 8PSK, QPSK) and reverses on frequency/analog modulations, consistent with a per-edge decision-boundary mechanism.

### C. Where the KAN advantage lives

To locate the KAN advantage we resolve accuracy by modulation on the mid-band (0 to 12 dB), the regime where the SNR sweep places the spline-KAN ahead of the MLP; this decomposes the high-SNR crossover, not the low-SNR headline of Table III, and we report it as such. In this regime (Fig. 4) the spline-KAN's advantage over the matched MLP is not spread evenly: it concentrates on the phase- and amplitude-shift modulations whose decision regions are constellation geometry, gaining $+0.23$ on BPSK, $+0.20$ on PAM4, and $+0.18$ on 8PSK, while it slightly trails the MLP on the analog and frequency-modulated classes (for example $-0.10$ on GFSK). This pattern is consistent with, though it does not prove, a per-edge mechanism: a learnable univariate function per input can carve the smooth amplitude/phase boundaries of a constellation, but offers little for signals whose class is carried by frequency dynamics that a flat model reads only weakly in any case. The KAN is thus not a generically better classifier; where it helps, it is a better fit to a specific, identifiable structure.

## VI. Regime 3: OFDM Channel Estimation

To test a third, structurally different task, we add OFDM channel estimation: an *estimation* problem rather than PA modeling or classification. On an $N{=}64$ subcarrier grid with $L{=}8$ multipath taps (exponential power-delay profile), comb pilots observe the channel frequency response at 16 subcarriers under noise at 10 dB SNR, and the task is to estimate the channel at all subcarriers. The metric is channel NMSE in dB, over 5 seeds. The decisive point is the baseline: linear minimum mean-square error (LMMSE) estimation is the Wiener estimator, optimal in the linear-Gaussian setting, and is the genuine classical workhorse here. From the noisy pilot observations $\mathbf{y}_p = \mathbf{H}_p + \mathbf{n}$ it forms

$$\hat{\mathbf{H}} = \mathbf{R}_{HH_p}\left(\mathbf{R}_{H_pH_p} + \sigma^2\mathbf{I}\right)^{-1}\mathbf{y}_p, \tag{9}$$

TABLE IV
OFDM channel estimation at 10 dB SNR, matched capacity, 5 seeds. NMSE in dB (lower is better).

| Model | NMSE [dB] ↓ |
|---|---|
| LMMSE (classical) | $\mathbf{-13.37 \pm 0.01}$ |
| MLP (matched) | $-13.28 \pm 0.01$ |
| KAN, spline (canon.) | $-13.28 \pm 0.01$ |
| KAN, Fourier | $-13.12 \pm 0.01$ |
| linear interpolation | $-9.48 \pm 0.01$ |

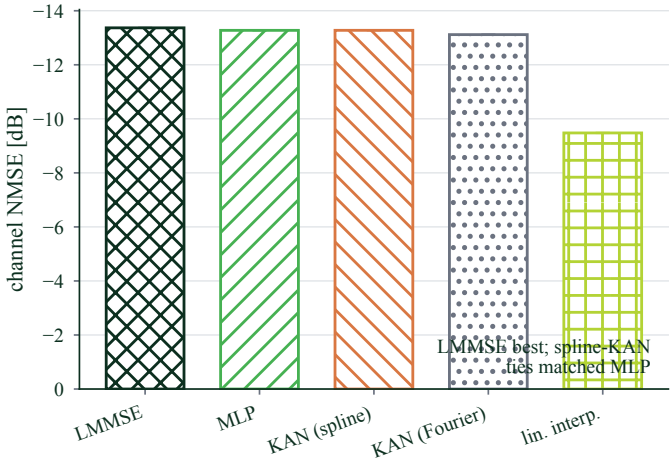



Fig. 5. Channel estimation, matched capacity. Classical LMMSE is the best estimator; the spline-KAN exactly ties the matched MLP and neither beats LMMSE, mirroring the PA-modeling regime on a different task type.

with $\mathbf{R}_{HH_p}$ and $\mathbf{R}_{H_pH_p}$ the channel cross- and pilot auto-covariances estimated from training; we also include simple linear interpolation as a weak reference.

Table IV and Fig. 5 report the result. The classical LMMSE is the best estimator at $-13.37$ dB, ahead of every neural model. Among the neural models the spline-KAN ($-13.28$ dB) exactly ties the matched MLP ($-13.28$ dB), the Fourier-KAN ($-13.12$ dB) is marginally behind, and all three clear the linear-interpolation reference ($-9.48$ dB) by a wide margin. As in the PA-modeling regime, the KAN buys nothing over a same-size MLP, and the right classical method is not beaten by either. Three regimes, three task types, and the same conclusion at the default configuration: KANs are competitive but not superior, and never beat the appropriate classical or structural baseline.

## VII. Interpretability Audit

The second KAN claim is interpretability: the learned univariate edges are simple enough to read off and replace by closed-form symbols, yielding an interpretable model "for free." A claim of this kind must satisfy two conditions, and we test them separately on the DPD KAN, where the task is smooth and thus most favorable to the claim.

*(Q1) Simplicity.* Are the learned first-layer edge functions well approximated by a human-readable low-order polynomial? We sample each edge on its input support and fit a cubic (over 5 seeds). The fits are excellent: the median per-edge $R^2$ is 0.998 and 99.7% of edges exceed $R^2 = 0.95$ (Fig. 7).




Fig. 6. Three representative learned KAN edge functions (solid) and their cubic fits (dashed), spanning the best, median, and worst per-edge fit. The fits are nearly coincident, confirming the edges are simple, readable curves. Note the near-flat median and worst-fit edges carry little of the model, which is why the variance-weighted $R^2$ is reported.

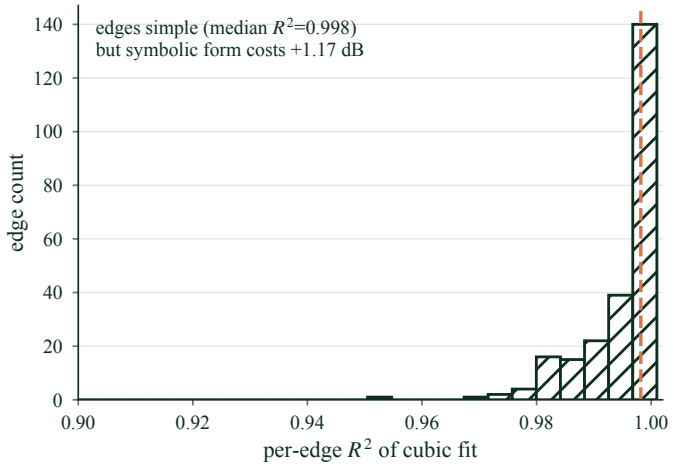



Fig. 7. Interpretability audit on the DPD KAN. The learned edges are simple (a cubic fits with median $R^2 = 0.998$), but distilling them into that symbolic form costs 1.17 dB of NMSE: the interpretable model is not the performant one.

To guard against near-flat edges inflating this, we also weight each edge's $R^2$ by its function variance; the variance-weighted mean is $0.998$, so the high-amplitude edges that actually carry the model are simple too, not just the negligible ones. Fig. 6 makes this concrete by overlaying three representative learned edges, spanning the best, median, and worst fits, on their cubics: the dashed fits sit essentially on top of the solid curves, and the highest-magnitude edge is a simple monotonic S-shape reminiscent of an AM/AM characteristic. The edges are indeed simple; the claim is not vacuous.

*(Q2) Fidelity: is it free?* We then *distill* the first layer into those cubics (a symbolic surrogate), keep the rest of the network exact, and re-measure NMSE over the same $5$ seeds. The surrogate degrades NMSE by $1.17 \pm 0.21$ dB. In other words, the readable form is not the performant form: extracting the interpretable symbols discards more than a decibel of modeling accuracy, consistently across seeds. Because we symbolize only the first layer, this is a *lower bound* on the cost of a fully symbolic KAN, and a task-loss-refit surrogate might recover part of it; even so, the naive extraction that the interpretability claim relies on is not free. This is precisely the nuance that qualitative "plot a few edges" demonstrations omit.

## VIII. Ablations: Stress-Testing the Comparison

A single matched-budget comparison could be unfair to KANs in ways that a proponent would rightly object to: perhaps the budget is too small, the KAN is under-tuned, the data too plentiful, or the operating point unrepresentative. We therefore run four ablations, each designed to give the KAN its best shot, and report the honest crossovers they reveal.

*Budget sweep (Fig. 8).* We sweep the DPD parameter budget from about $0.8$k to $9$k. The MLP-over-KAN ordering persists at every budget, and the gap does not close: the Fourier-KAN improves with budget but tracks below the MLP, and the spline-KAN plateaus near $-14$ dB and stops improving.

TABLE V
SYMMETRIC MLP SWEEP ON DPD AT THE DEFAULT BUDGET (5 SEEDS). A TUNED THREE-LAYER MLP IS THE STRONGEST MATCHED-BUDGET MODEL, SO THE KAN IS NOT BEATEN BY AN UNTUNED BASELINE. THE DEFAULT 2-LAYER ROW ($-16.25$) IS THIS SWEEP'S OWN RUN; IT MATCHES TABLE II'S $-16.27$ WITHIN SEED NOISE.

| MLP configuration | Params | NMSE [dB] ↓ |
|---|---|---|
| 1 layer, ReLU | 2329 | $-15.22 \pm 0.14$ |
| 2 layer, ReLU (default) | 2354 | $-16.25 \pm 0.08$ |
| 2 layer, GELU | 2354 | $-14.48 \pm 0.22$ |
| 2 layer, tanh | 2354 | $-15.78 \pm 0.13$ |
| 3 layer, ReLU | 2389 | $\mathbf{-16.72 \pm 0.17}$ |
| 4 layer, ReLU | 2277 | $-16.60 \pm 0.10$ |

Giving the KAN more parameters of the same kind does not rescue it here.

*KAN and MLP hyperparameters (Tables VI and V).* Fairness demands that if we tune the KAN, we tune the MLP too. We do both. Increasing the Fourier basis order to $K = 8$ raises the Fourier-KAN to $-17.14$ dB at $4634$ parameters, and spectral richness, not width, is the lever. But the MLP is not fixed either: sweeping its depth and activation (Table V), a three-layer MLP reaches $-16.72$ dB at the default budget, beating every matched-budget KAN there, and $-16.95$ dB at the larger $K = 8$ budget. So the Fourier-KAN's regression win survives a *symmetrically* tuned MLP, but only by $0.19$ dB, down from $0.6$ dB against the untuned MLP: a real but thin advantage, and only at the larger budget. This is the flip side of the capacity-confound argument: just as an unmatched KAN can win for the wrong reason, a well-chosen KAN can win for the right one, and only a matched, symmetrically tuned comparison tells them apart.

*Sample efficiency (Fig. 9).* A distinct claimed KAN strength is data efficiency. It holds here: with $1$k and $4$k training examples the Fourier-KAN beats the matched MLP (paired differences $-0.23$ and $-0.37$ dB, intervals excluding zero), and the ordering inverts only once data is plentiful ($+0.76$ dB at the full set). A KAN is a reasonable choice when labeled data is the binding constraint.

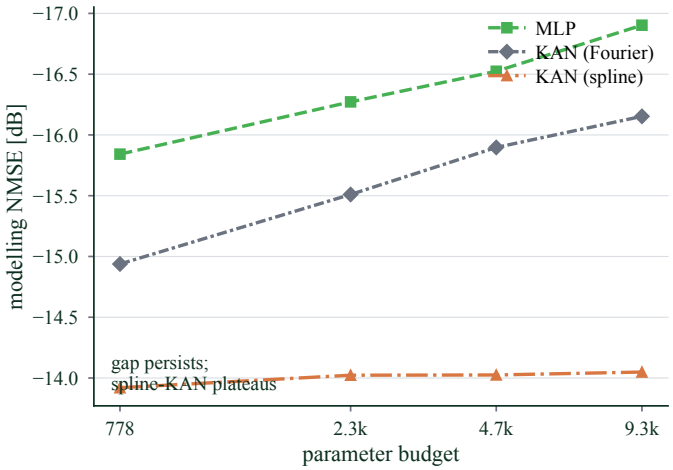



Fig. 8. DPD budget sweep. The MLP leads at every budget; the Fourier-KAN improves but tracks below it, and the spline-KAN plateaus. More same-kind parameters do not close the gap.

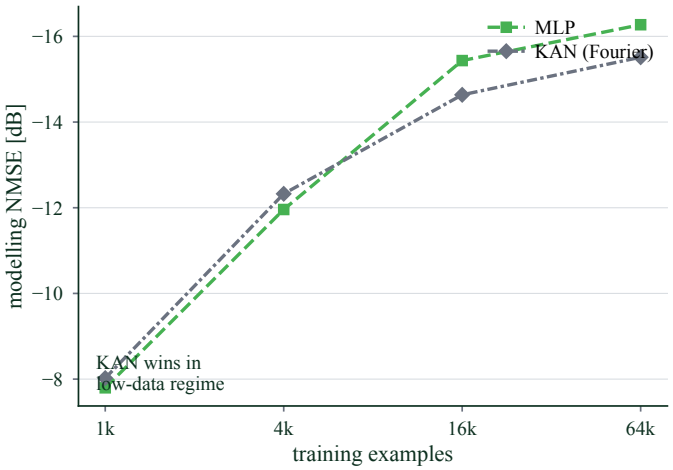



Fig. 9. DPD sample efficiency. The Fourier-KAN beats the matched MLP when training data is scarce ($\leq$ 4k examples); the ordering inverts only once data is plentiful.

TABLE VI
DPD KAN HYPERPARAMETER SWEEP (5 SEEDS). A RICHER FOURIER BASIS ($K$=8, −17.14 DB AT 4634 PARAMS) BEATS EVEN A SYMMETRICALLY TUNED MLP (−16.95 DB AT THAT BUDGET), BUT BY ONLY 0.19 DB (SEE TABLE V).

| Variant | Setting | Params | NMSE [dB] ↓ |
|---|---|---|---|
| Fourier | $K = 2$ | 1178 | $-14.42 \pm 0.05$ |
| Fourier | $K = 4$ | 2330 | $-15.54 \pm 0.09$ |
| Fourier | $K = 8$ | 4634 | $\mathbf{-17.14 \pm 0.12}$ |
| spline | grid 4 | 1152 | $-13.03 \pm 0.15$ |
| spline | grid 6 | 1440 | $-14.04 \pm 0.13$ |
| spline | grid 10 | 2016 | $-15.09 \pm 0.09$ |

*SNR sweep (Fig. 10).* Finally we resolve the AMC accuracy by SNR instead of aggregating over a band. The convolutional baseline leads at every SNR. Among the flat models, the MLP is best only at low SNR; the spline-KAN overtakes it above about 2 dB (reaching 0.620 versus the MLP's 0.480 at 10 dB), and over the *full* SNR range the spline-KAN is the best flat model (0.375 versus 0.333). The aggregate ranking is thus an artifact of which band one reports, which is the saturated-benchmark problem in a subtler form: not only can a band be too easy, it can be chosen (even inadvertently) to favor one model.

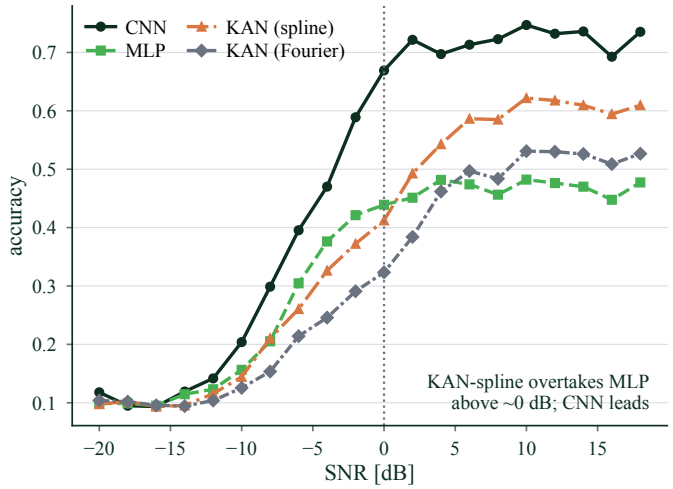



Fig. 10. AMC accuracy resolved by SNR. The CNN leads throughout; among flat models the MLP wins only at low SNR, while the spline-KAN overtakes it above $\sim$ 2 dB. Reporting a single band hides this crossover.

## IX. COMPUTATIONAL COST: THE EFFICIENCY ILLUSION

Matching parameter count, the axis on which KAN "efficiency" is usually claimed, hides the axis that matters for deployment: compute. A KAN edge evaluates a spline or a truncated Fourier series where an MLP edge evaluates a single multiply-add, so at equal parameters a KAN does substantially more work. Counting multiply-adds per layer,

$$C_{\mathrm{MLP}} = 2\, d_{\mathrm{in}} d_{\mathrm{out}}, \tag{10}$$

$$C^{\mathrm{F}}_{\mathrm{KAN}} = 2K\, d_{\mathrm{in}} d_{\mathrm{out}} + \kappa\,(2K)\, d_{\mathrm{in}}, \tag{11}$$

$$C^{\mathrm{S}}_{\mathrm{KAN}} = (G{+}k)\, d_{\mathrm{in}} d_{\mathrm{out}} + \rho\, d_{\mathrm{in}}, \tag{12}$$

where $\kappa$ is the cost of a $\cos / \sin$ pair and $\rho$ that of the B-spline recurrence. Two effects compound: the basis multiplies the connectivity term by $2K$ or $G{+}k$, and the transcendental or recurrence terms add fixed per-edge overhead that no parameter count reflects. The parameter budget absorbs the multiplicative factor by shrinking the hidden width, but the per-edge overhead does not shrink with it; it reappears as latency.

Table VII measures it. At the *same* parameter budget, inference is 7× to 30× slower for the KANs than for the matched MLP on the regression models, and 4.5× to 21× slower on the classification models, with the canonical spline-KAN the most expensive by far. So even in the regimes where a KAN edges out the MLP on accuracy, it pays with an order of magnitude more compute per inference, and a practitioner optimizing performance per FLOP or per millisecond, the realistic constraint at a base station or on an edge device, would not choose it. "Fewer parameters" is the wrong metric; matched-parameter is already generous to the KAN. These timings are single-thread CPU measurements; on a massively

TABLE VII
COMPUTE AT MATCHED PARAMETERS. INFERENCE TIME PER BATCH (256) AND SLOWDOWN VERSUS THE MATCHED MLP. KANS COST FAR MORE COMPUTE FOR THE SAME PARAMETER COUNT.

| Task | Model | Params | Infer. [ms] ↓ |
|---|---|---|---|
| DPD | MLP | 2354 | 0.04 |
| DPD | KAN, Fourier | 2330 | 0.27 |
| DPD | KAN, spline | 2280 | 1.14 |
| AMC | CNN | 5291 | 25.2 |
| AMC | MLP | 68651 | 0.16 |
| AMC | KAN, Fourier | 68395 | 0.74 |
| AMC | KAN, spline | 69420 | 3.47 |

parallel accelerator the wall-clock gap would narrow, since the extra per-edge operations parallelize. The *arithmetic*, however, does not: the multiply-add and transcendental counts (10)–(12) are device-independent, so the energy and throughput penalty remains even where latency is hidden. We report CPU latency as the transparent proxy and the FLOP counts as the device-independent statement.

## X. DISCUSSION

### A. A regime map

Combining the regimes, the audit, and the ablations yields a compact map with three tiers. *Where KANs help:* at higher SNR in modulation classification, over the full SNR range, in the low-data regime, and, with a richer basis, on smooth regression at matched capacity, a KAN is the better flat model, sometimes by a clear margin. Their learnable edges are a real tool, not a gimmick. *Where they do not:* at the default budget and basis, at low SNR, and with plentiful data, a same-size MLP is as good or better, so a KAN advantage is never automatic. *Where architecture matters more than either:* on structured I/Q classification a small CNN with the right inductive bias beats every flat model by a wide margin at a fraction of the parameters, so if the task has exploitable structure, the budget is better spent on the prior than on the edge parameterization. Layered on top, KAN interpretability is real but not free: the edges are genuinely simple, yet the symbolic extraction that would make them interpretable costs more than a decibel.

### B. Interpreting the crossovers

The crossovers are not random, and two mechanisms are consistent with all our observations, though we frame them as hypotheses rather than claims. First, the KAN's per-edge nonlinearity is most useful when the target relationship is a clean low-dimensional function of each input. At high SNR the received constellation is sharp, so the mapping from I/Q samples to class is close to a smooth geometric partition that per-edge functions can carve; at low SNR the same flexibility fits noise, which is exactly where the more constrained MLP, and far more so the translation-equivariant CNN, generalize better. This matches both the SNR crossover and the low-data crossover: fewer, cleaner examples favor the model whose inductive bias is a smooth univariate decomposition. Second, on the smooth regression task the benefit of the Fourier basis grows with its order $K$ rather than with width, because a power-amplifier AM/AM curve is well represented by a few harmonics; this is why $K{=}8$ overtakes the MLP while wider low-$K$ KANs do not. Neither mechanism supports a blanket advantage; both predict a bounded regime, which is what we measure.

### C. Why the literature reports sweeping wins

Our results both confirm that KANs can win and explain why the reported wins are overstated. A KAN with an unmatched budget, or with a richer basis than the MLP it is compared to, will beat that MLP, and our own ablations reproduce exactly such a win once the KAN is tuned; attributing it to "the architecture" rather than to the extra effective capacity is the error. Likewise, a saturated benchmark makes every model tie so any fluctuation reads as a win, and an aggregate over a favorable band hides the crossover we measured. The advantage does not vanish under matched control, but it shrinks to a specific, honest set of regimes rather than the blanket superiority usually claimed.

### D. Threats to validity

We tested three regimes, two KAN variants, and one dataset per regime; a broader sweep (more datasets, wavelet or Chebyshev KANs [12], [14], larger budgets) would refine the regime boundaries, and we release the harness to make that cheap. The DPD PA is simulated, though from a standard non-polynomial model, so the classical baseline is not privileged; a measured PA would strengthen the regression result. The AMC study is capped in sample count for tractability. Because our conclusions are two-sided, the main risk is mislocating a boundary rather than a hidden global winner: every reported crossover has intervals that exclude zero, and the interpretability cost was measured directly rather than inferred.

### E. Future work

Three extensions would sharpen the map. First, a *systematic audit* of the recent KAN-in-wireless literature against the six-point checklist of Section X: our results predict that the reported efficiency wins will correlate with unmatched capacity, saturated benchmarks, or unreported seeds, and a structured survey scoring each paper on the six criteria would test that prediction directly and quantify how widespread the confounds are. Second, a full DPD linearization loop with ACPR and EVM on a *measured* power amplifier, replacing our forward Saleh model, would confirm whether the thin regression crossover survives real memory and hysteresis effects. Third, extending the model set to wavelet and Chebyshev KANs [12], [14] and adding a real multimodal benchmark such as DeepSense 6G [36] would probe whether any KAN variant reverses the ordering on a task with genuinely exploitable structure. The released harness is designed to make each of these a small addition rather than a reimplementation.

### F. Practical guidance

For a wireless practitioner: if the task has exploitable structure, spend the model budget on the right inductive bias (a convolution, an unrolled algorithm) rather than on the edge parameterization, which dominated every flat model here. Among flat models, reach for a KAN when labeled data is scarce or when the operating regime is clean (high SNR), where it beat the matched MLP, and tune its basis richness, not just its width; otherwise a same-size MLP is a strong and simpler default. If interpretability is the goal, measure the accuracy cost of the symbolic extraction rather than assuming it is free. The recurring discipline is to match capacity and resolve the operating point before believing any architecture comparison.

### G. A reporting checklist

The confounds we removed are cheap to check, and we distill them into a checklist a reviewer or author can apply to any KAN-versus-baseline wireless claim:

1) **Matched capacity.** Are the KAN and its baseline sized to the same parameter count, reported explicitly? An efficiency claim against a smaller-parameter baseline is not evidence about the architecture.
2) **Non-saturated benchmark.** Is the reported operating point one where models are below the ceiling? If every model exceeds $\sim 99\%$, the comparison is uninformative.
3) **Resolved operating point.** Is accuracy resolved by SNR (or the relevant axis) rather than aggregated over a band that may favor one model, as our crossover shows?
4) **Multiple seeds with intervals.** Is the claim supported by a paired difference with a confidence interval that excludes zero, not a single-seed number?
5) **Strong baseline.** Is the baseline the one a practitioner would use (a CNN for I/Q, a memory polynomial for DPD), not a deliberately weak MLP?
6) **Interpretability with fidelity.** If interpretability is claimed, is the accuracy cost of the symbolic extraction measured, not just a plot of a few edges?

Every "win" we could not reproduce under matched control failed at least one of these checks; every genuine KAN win we found survives all six.

## XI. Reproducibility

Every number in this paper is produced by a script and checked against a machine-readable claims file. The harness (`harness.py`) implements the KAN variants, the MLP, the matched-capacity sizing, the training loop, and the bootstrap statistics; `exp_dpd.py`, `exp_amc.py`, and `exp_interp.py` produce the main tables and Figs. 2–7, and `abl_dpd.py` and `abl_amc.py` produce the ablations (Table VI, Figs. 8–10), each from its own result CSV. The RML2016.10a dataset is public [28]. We list, for each table and figure, the exact script and CSV that generate it, and we fix all random seeds. The code and a `claims.json` single-source-of-truth are released so that a reader can regenerate the results and verify the paper's numbers against the CSVs. The full code, result CSVs, and the claims file are available at https://github.com/haodpsut/kan-wireless-benchmark.

## XII. Conclusion

We asked when Kolmogorov–Arnold networks help in wireless, and answered by measurement. Under a parameter-matched, multi-seed, non-saturated protocol, the honest answer is two-sided: KANs are competitive and win in specific regimes, at higher SNR in modulation classification, in the low-data regime, and with a richer basis on smooth regression, but they never dominate, are beaten decisively by the right inductive bias on structured tasks, lose to a same-size MLP at the default budget and at low SNR, and carry an interpretability cost of more than a decibel to extract. The sweeping "efficiency wins" reported in the wireless KAN literature are consistent with unmatched capacity and saturated or favorably chosen benchmarks rather than with a blanket architectural advantage. We hope the regime map, the protocol, and the released harness make the next such claim easier to place, and to check, before it is believed.

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